

1. Handling Missing Values

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	
07-JUN-UK	Headphones	Sony		45	Electroni
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	
08-DEC-DE	Coffee Maker	Nespresso	120	35	
18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni
16-APR-ES	headphones	Bose	250	20	Electroni
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electroni
27-JAN-IT	laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman		10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion
11-JUL-CA	Blender	Vitamix	400	40	Kitchen
16-APR-ES	headphones	Bose	250	20	Electroni
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronics
09-JUL-FR	Watch	Casio	100	20	Accessories

Not Correction

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	UNKNOWN
07-JUN-UK	Headphones	Sony	309	45	Electroni
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	UNKNOWN
08-DEC-DE	Coffee Maker	Nespresso	120	35	UNKNOWN
18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni
16-APR-ES	headphones	Bose	250	20	Electroni
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electroni
27-JAN-IT	laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	309	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	UNKNOWN
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	309	25	Fashion
11-JUL-CA	Blender	Vitamix	400	40	Kitchen
16-APR-ES	headphones	Bose	250	20	Electroni
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronics
09-JUL-FR	Watch	Casio	100	20	Accessories

Correction

- After the correction data Price filling in the blank used **“If”** function and **“Isblank”** but the true value is average of the Price function then category also filled the blank used same function but the true value is **“UNKNOWN”**.

2. Correcting Inconsistent Data

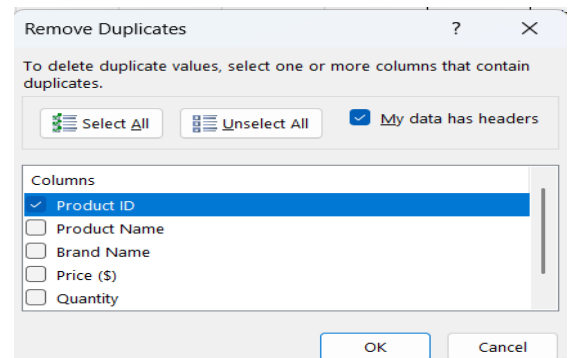
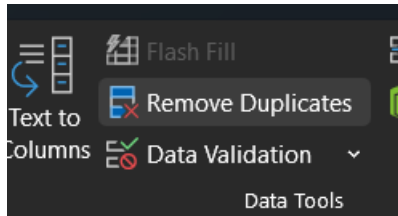
Product Name	Category		Pro-correction	cat-correction
laptop	Electronics		Laptop	Electronics
Sneakers	Fashion		Sneakers	Fashion
Coffee Maker	Kitchen		Coffee Maker	Kitchen
smartphone	Electronics		Smartphone	Electronics
Backpack	UNKNOWN		Backpack	UNKNOWN
Headphones	Electroni		Headphones	Electronics
T-shirt	Fashion		T-Shirt	Fashion
Blender	Kitchen		Blender	Kitchen
Tablet	Electronics		Tablet	Electronics
Hiking Boots	Outdoor		Hiking Boots	Outdoor
Laptop	Electronics		Laptop	Electronics
Sneakers	UNKNOWN		Sneakers	UNKNOWN
Coffee Maker	UNKNOWN		Coffee Maker	UNKNOWN
Smartwatch	Electroni		Smartwatch	Electronics
headphones	Electroni		Headphones	Electronics
Laptop Bag	Accessories		Laptop Bag	Accessories
Smartwatch	Electroni		Smartwatch	Electronics
laptop	Electronics		Laptop	Electronics
Sunglasses	Fashion		Sunglasses	Fashion
Camping Tent	Outdoor		Camping Tent	Outdoor
Camera	Electronics		Camera	Electronics
Microwave	Kitchen		Microwave	Kitchen
Fitness Tracker	UNKNOWN		Fitness Tracker	UNKNOWN
Laptop Bag	Accessories		Laptop Bag	Accessories
Smartphone	Electronics		Smartphone	Electronics
Sunglasses	Fashion		Sunglasses	Fashion
Blender	Kitchen		Blender	Kitchen
headphones	Electroni		Headphones	Electronics
Dress	Fashion		Dress	Fashion
Toaster	Kitchen		Toaster	Kitchen
Fitness Tracker	Electronics		Fitness Tracker	Electronics
Jeans	Fashion		Jeans	Fashion
Laptop	Electronics		Laptop	Electronics
Watch	Accessories		Watch	Accessories

Before

After

- ❖ This data not clean so Product name is inconsistent so used Function is **“PROPER”** and then category is Name is not correct format so using filter to change the category name and 3rd step is this two name also change to **“Text Format”**

3. Duplicate Removal Process



1. Select Removal duplicates

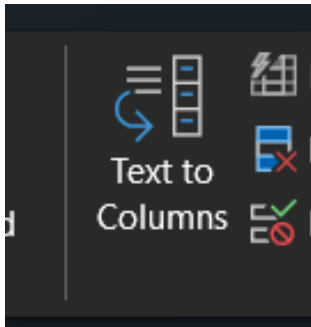
2. Select Product ID



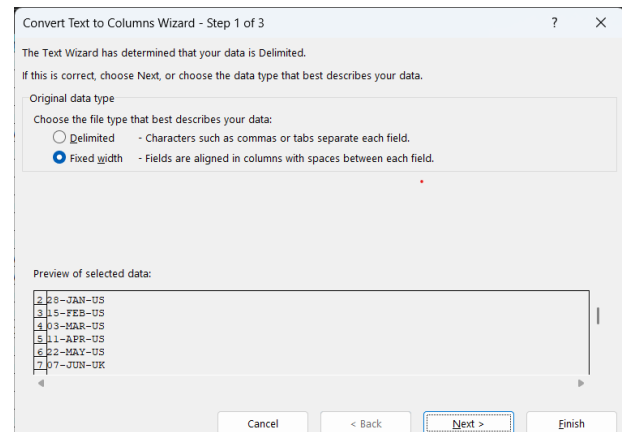
3. Result (Duplicate remove)

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	UNKNOWN
07-JUN-UK	Headphones	Sony	309	45	Electronics
19-JUL-UK	T-Shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	UNKNOWN
08-DEC-DE	Coffee Maker	Nespresso	120	35	UNKNOWN
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	309	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	UNKNOWN
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	309	25	Fashion
11-JUL-CA	Blender	Vitamix	400	40	Kitchen
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
09-JUL-FR	Watch	Casio	100	20	Accessories

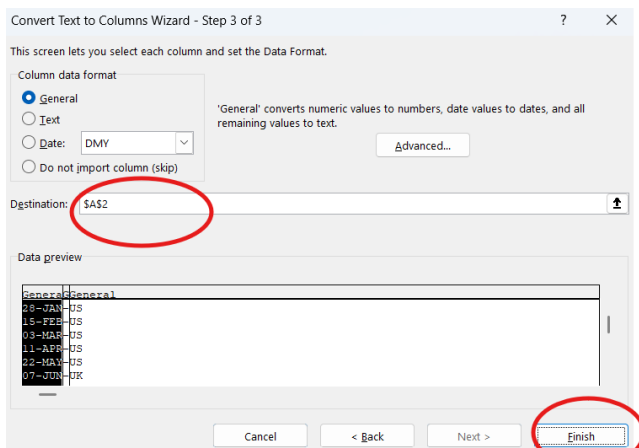
4. Splitting and Merging Data



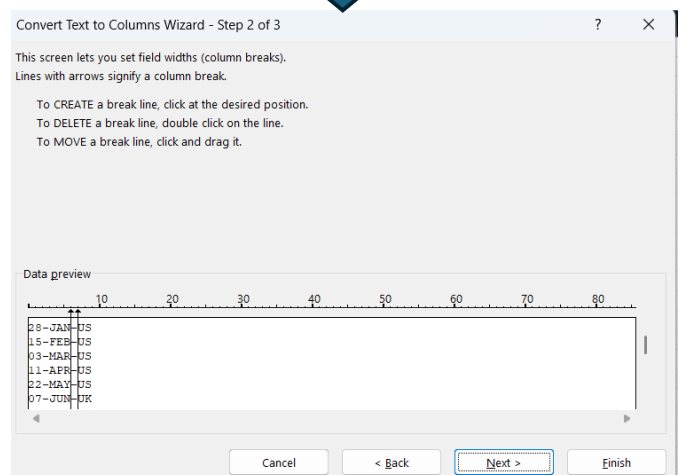
1. Select data and click text to column



2. Select Fixed with and click next



4. Select destination column and then finish



3. Separate Date and country and click next

Product	t	ID
28-Jan	-	US
15-Feb	-	US
03-Mar	-	US
11-Apr	-	US
22-May	-	US
07-Jun	-	UK
19-Jul	-	UK
23-Aug	-	UK
05-Sep	-	UK
14-Oct	-	UK
17-Jun	-	IN
25-Nov	-	AU
08-Dec	-	DE
18-Feb	-	CA
16-Apr	-	ES
21-Aug	-	CA
20-Aug	-	CN
27-Jan	-	IT
01-Mar	-	UK
14-Aug	-	US
14-May	-	RU
09-Jan	-	CA
19-Jul	-	BR
29-Sep	-	CA
03-Jun	-	CA
11-Jul	-	CA
07-Mar	-	CA
13-Apr	-	CA
24-May	-	CA
02-Dec	-	CA
09-Jul	-	FR

5. Final Separate into two column

Merging

Product Name	Brand Name
Laptop	Dell
Sneakers	Nike
Coffee Maker	Keurig
Smartphone	Samsung
Backpack	North Face
Headphones	Sony
T-Shirt	Adidas
Blender	Ninja
Tablet	Apple
Hiking Boots	Timberland
Laptop	HP
Sneakers	Adidas
Coffee Maker	Nespresso
Smartwatch	Fitbit
Headphones	Bose
Laptop Bag	Samsonite
Smartwatch	Huawei
Laptop	Asus
Sunglasses	Oakley
Camping Tent	Coleman
Camera	Nikon
Microwave	Panasonic
Fitness Tracker	Xiaomi
Smartphone	Google
Sunglasses	Ray-Ban
Blender	Vitamix
Dress	Zara
Toaster	Hamilton
Fitness Tracker	Garmin
Jeans	Levi's
Watch	Casio

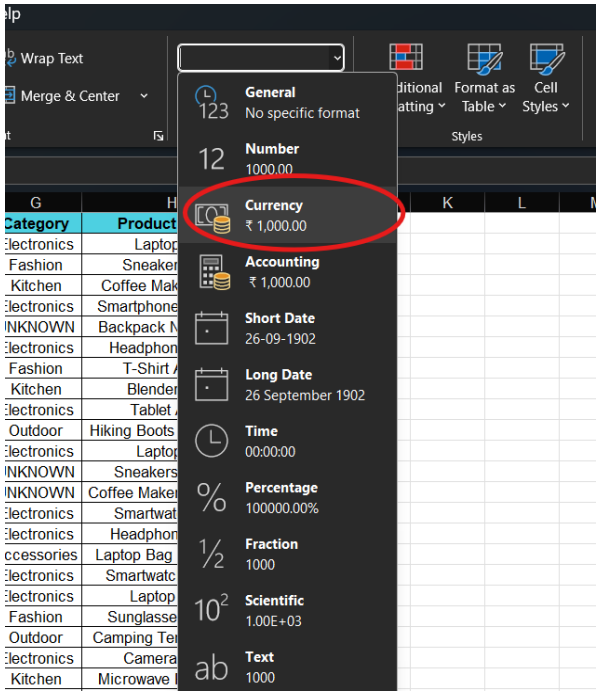
Before

product brand
Laptop Dell
Sneakers Nike
Coffee Maker Keurig
Smartphone Samsung
Backpack North Face
Headphones Sony
T-Shirt Adidas
Blender Ninja
Tablet Apple
Hiking Boots Timberland
Laptop HP
Sneakers Adidas
Coffee Maker Nespresso
Smartwatch Fitbit
Headphones Bose
Laptop Bag Samsonite
Smartwatch Huawei
Laptop Asus
Sunglasses Oakley
Camping Tent Coleman
Camera Nikon
Microwave Panasonic
Fitness Tracker Xiaomi
Smartphone Google
Sunglasses Ray-Ban
Blender Vitamix
Dress Zara
Toaster Hamilton
Fitness Tracker Garmin
Jeans Levi's
Watch Casio

After

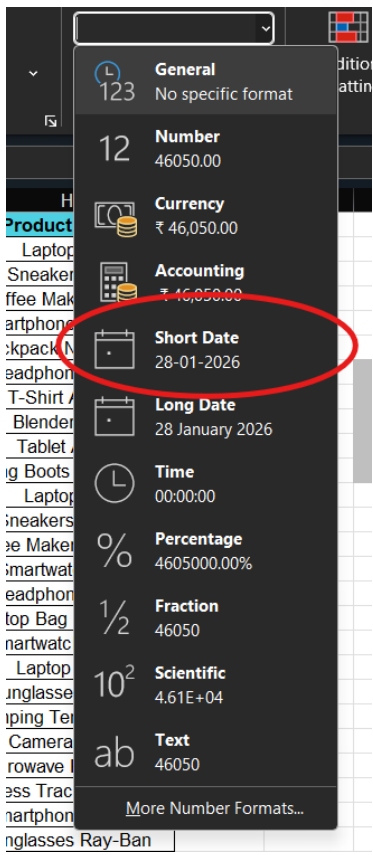
- ❖ This data is used in Merging function so used the function is “**CONCATENATE**” to merging this data so this data name is Product Brand.

5. Number Formatting



1. Select data and then click currency

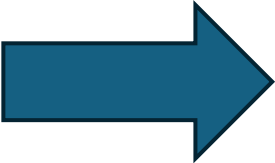
Price (\$)
₹ 1,000.00
₹ 80.00
₹ 130.00
₹ 900.00
₹ 70.00
₹ 309.00
₹ 30.00
₹ 90.00
₹ 500.00
₹ 130.00
₹ 950.00
₹ 90.00
₹ 120.00
₹ 150.00
₹ 250.00
₹ 50.00
₹ 160.00
₹ 980.00
₹ 150.00
₹ 309.00
₹ 700.00
₹ 80.00
₹ 150.00
₹ 800.00
₹ 309.00
₹ 400.00
₹ 60.00
₹ 40.00
₹ 130.00
₹ 50.00
₹ 100.00



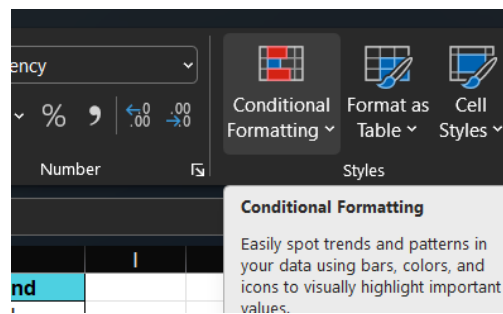
Finally convert to currency value

Manufacturing Date
28-01-2026
15-02-2026
03-03-2026
11-04-2026
22-05-2026
07-06-2026
19-07-2026
23-08-2026
05-09-2026
14-10-2026
17-06-2026
25-11-2026
08-12-2026
18-02-2026
16-04-2026
21-08-2026
20-08-2026
27-01-2026
01-03-2026
14-08-2026
14-05-2026
09-01-2026
19-07-2026
29-09-2026
03-06-2026
11-07-2026
07-03-2026
13-04-2026
24-05-2026
02-12-2026
09-07-2026

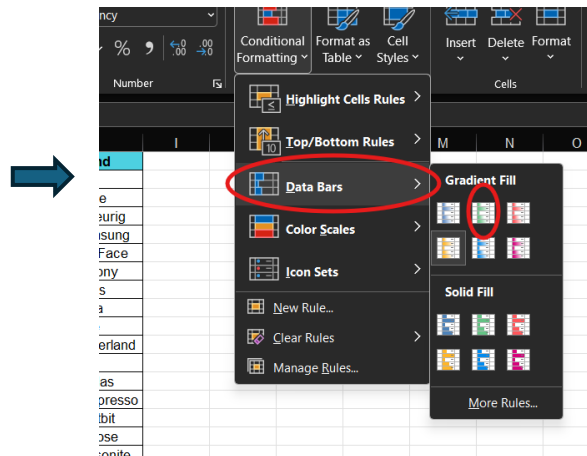
Finally convert to date value



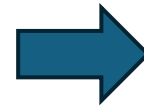
6. Conditional Formatting



1. Select Data and then click Conditional Formatting

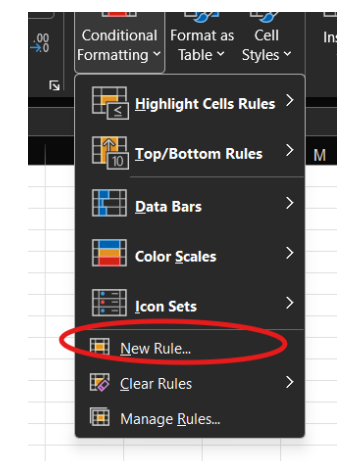


2. Then select Data Bar and then click colour

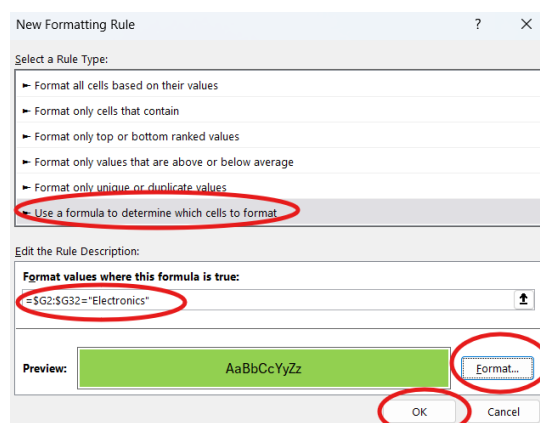


Price (\$)
₹ 1,000.00
₹ 80.00
₹ 130.00
₹ 900.00
₹ 70.00
₹ 309.00
₹ 30.00
₹ 90.00
₹ 500.00
₹ 130.00
₹ 950.00
₹ 90.00
₹ 120.00
₹ 309.00
₹ 150.00
₹ 250.00
₹ 50.00
₹ 160.00
₹ 980.00
₹ 150.00
₹ 309.00
₹ 700.00
₹ 80.00
₹ 150.00
₹ 800.00
₹ 309.00
₹ 400.00
₹ 60.00
₹ 40.00
₹ 130.00
₹ 50.00
₹ 100.00

3. Result price list change the colour



1. select data and then click conditional formatting and then click New rule



2. Click use a formula and then type formula and then click format in colour and then ok



Category
Electronics
Fashion
Kitchen
Electronics
UNKNOWN
Electronics
Fashion
Kitchen
Electronics
Outdoor
Electronics
UNKNOWN
UNKNOWN
Electronics
Electronics
Accessories
Electronics
Electronics
Fashion
Outdoor
Electronics
Kitchen
UNKNOWN
Electronics
Fashion
Kitchen
Fashion
Kitchen
Electronics
Fashion
Accessories

3. Result Electronics item change the colour

RESULT

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	
07-JUN-UK	Headphones	Sony		45	Electroni
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	
08-DEC-DE	Coffee Maker	Nespresso	120	35	
18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni
16-APR-ES	headphones	Bose	250	20	Electroni
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electroni
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman		10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
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11-JUL-CA	Blender	Vitamix	400	40	Kitchen
16-APR-ES	headphones	Bose	250	20	Electroni
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronics
09-JUL-FR	Watch	Casio	100	20	Accessories

BEFORE
CLEANING

Manufacturing Date	County Code	Product Name	Brand Name	Price (\$)	Quantity	Category	Product
28-01-2026	US	Laptop	Dell	₹ 1,000.00	30	Electronics	Laptop Dell
15-02-2026	US	Sneakers	Nike	₹ 80.00	15	Fashion	Sneakers Nike
03-03-2026	US	Coffee Maker	Keurig	₹ 130.00	40	Kitchen	Coffee Maker Keurig
11-04-2026	US	Smartphone	Samsung	₹ 900.00	25	Electronics	Smartphone Samsung
22-05-2026	US	Backpack	North Face	₹ 70.00	20	UNKNOWN	Backpack North Face
07-06-2026	UK	Headphones	Sony	₹ 309.00	45	Electronics	Headphones Sony
19-07-2026	UK	T-Shirt	Adidas	₹ 30.00	5	Fashion	T-Shirt Adidas
23-08-2026	UK	Blender	Ninja	₹ 90.00	35	Kitchen	Blender Ninja
05-09-2026	UK	Tablet	Apple	₹ 500.00	50	Electronics	Tablet Apple
14-10-2026	UK	Hiking Boots	Timberland	₹ 130.00	10	Outdoor	Hiking Boots Timberland
17-06-2026	IN	Laptop	HP	₹ 950.00	25	Electronics	Laptop HP
25-11-2026	AU	Sneakers	Adidas	₹ 90.00	40	UNKNOWN	Sneakers Adidas
08-12-2026	DE	Coffee Maker	Nespresso	₹ 120.00	35	UNKNOWN	Coffee Maker Nespresso
18-02-2026	CA	Smartwatch	Fitbit	₹ 150.00	15	Electronics	Smartwatch Fitbit
16-04-2026	ES	Headphones	Bose	₹ 250.00	20	Electronics	Headphones Bose
21-08-2026	CA	Laptop Bag	Samsonite	₹ 50.00	35	Accessories	Laptop Bag Samsonite
20-08-2026	CN	Smartwatch	Huawei	₹ 160.00	15	Electronics	Smartwatch Huawei
27-01-2026	IT	Laptop	Asus	₹ 980.00	10	Electronics	Laptop Asus
01-03-2026	UK	Sunglasses	Oakley	₹ 150.00	15	Fashion	Sunglasses Oakley
14-08-2026	US	Camping Tent	Coleman	₹ 309.00	10	Outdoor	Camping Tent Coleman
14-05-2026	RU	Camera	Nikon	₹ 700.00	50	Electronics	Camera Nikon
09-01-2026	CA	Microwave	Panasonic	₹ 80.00	20	Kitchen	Microwave Panasonic
19-07-2026	BR	Fitness Tracker	Xiaomi	₹ 150.00	30	UNKNOWN	Fitness Tracker Xiaomi
29-09-2026	CA	Smartphone	Google	₹ 800.00	45	Electronics	Smartphone Google
03-06-2026	CA	Sunglasses	Ray-Ban	₹ 309.00	25	Fashion	Sunglasses Ray-Ban
11-07-2026	CA	Blender	Vitamix	₹ 400.00	40	Kitchen	Blender Vitamix
07-03-2026	CA	Dress	Zara	₹ 60.00	30	Fashion	Dress Zara
13-04-2026	CA	Toaster	Hamilton	₹ 40.00	10	Kitchen	Toaster Hamilton
24-05-2026	CA	Fitness Tracker	Garmin	₹ 130.00	5	Electronics	Fitness Tracker Garmin
02-12-2026	CA	Jeans	Levi's	₹ 50.00	50	Fashion	Jeans Levi's
09-07-2026	FR	Watch	Casio	₹ 100.00	20	Accessories	Watch Casio

AFTER
CLEANING